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INFORMATION Duke University
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EDUCATION **Duke University, Ph.D.** (2018–Present)

- Computer Engineering: Deep Learning. GPA: 3.97/4.0
- Thesis topic: Continual Learning in Deep Neural Networks
- Advisor: Dr. Lawrence Carin

Delhi Technological University, B.Tech. (2010–2014)

- Degree: Software Engineering, First Class
- Department: Computer Science and Engineering

PREPRINTS • V. Verma, K. Liang, **N. Mehta**, & L. Carin, “Meta-Learned Attribute Self-Gating for Continual Generalized Zero-Shot Learning”. [\[preprint\]](#)

• V. Verma, **N. Mehta**, S. Si & L. Carin, “Pushing the Efficiency Limit using Structured Sparse Convolutions” (**New paper**) [\[preprint\]](#)

• W. Hao, **N. Mehta**, K. Liang, P. Cheng, M. Khamy & L. Carin, “WAFFLe: Weight Anonymized Factors for Federated Learning” [\[preprint\]](#)

PUBLICATIONS • **N. Mehta**, K. Liang, V. Verma & L. Carin, “Continual Learning using a Bayesian Nonparametric Dictionary of Weight Factors”. International Conference of Artificial Intelligence and Statistics (**AISTATS**), 2021. [\[pdf\]](#)

• **N. Mehta**, L. Carin & P. Rai, “Stochastic Blockmodels meet Graph Neural Networks”. International Conference on Machine Learning (**ICML**) 2019. [\[pdf\]](#)

• **N. Mehta**^{*}, J. Liu^{*}, C. Tao & L. Carin, “Estimation and Sampling of Unnormalized Statistical Models with Stein Score Matching”. Workshop on Stein’s Method at International Conference on Machine Learning (**ICML**) 2019. [\[pdf\]](#) [\[poster\]](#)

• V. Verma, K. Liang, **N. Mehta**, P. Rai, & L. Carin, “Efficient Feature Transform for Continual Learning”. Computer Vision and Pattern Recognition (**CVPR**), 2021. [\[pdf\]](#)

• S. Assaad, S. Zeng, C. Tao, S. Datta, **N. Mehta**, R. Henao, F. Li and L. Carin, “Counterfactual Representation Learning with Balancing Weights”. International Conference of Artificial Intelligence and Statistics (**AISTATS**), 2021. [\[pdf\]](#)

• A. Sarkar^{*}, **N. Mehta**^{*} & P. Rai, “Learning interpretable hierarchical node representations via Ladder Gamma VAE”. Workshop on Graph Representation Learning, Neural Information Processing Systems (**NeurIPS**) 2019. Full version of this work appeared at the Advancements of Artificial Intelligence (**AAAI**) 2020. [\[pdf\]](#)

• R. Panda, A. Pensia, **N. Mehta**, M. Zhou & P. Rai, “Deep Topic Models for Multi-label Learning”. International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2019. [\[pdf\]](#)

• **N. Mehta**, J. McBride & G. Pandey, “Robust and Fast 3D Scan Alignment using Mutual Information”. International Conference on Robotics and Automation (**ICRA**) 2018. [\[pdf\]](#)

• P. Chapfuwa, C. Li, **N. Mehta**, L. Carin and R. Henao, “Survival Cluster Analysis”, ACM Conference on Health, Inference, and Learning (**CHIL**) [\[pdf\]](#)

^{*} Equal Contribution

RESEARCH
EXPERIENCE

Google, Research Intern in Brain Team

(May 2021–Present)

- **Multiple User Representations for Recommendations:** Proposed a novel technique to generate multi-resolution representations for users that capture diverse interests leading to better recommendation quality. We show that learning multiple representations for users achieve a well-clustered output embedding space, in contrast to the embeddings learned by models having a single representation. Moreover, our work attempts to identify the effect of user characteristics (*e.g.* interest volatility, number of recent interests, *etc*) on the performance of the model.

Google, Research Intern in AutoML Team

(May 2020–August 2020)

- **One-shot Meta-Learning for Neural Networks in Auto-ML pipelines:** Proposed an automated meta-learning pipeline that offers a “good initialization” of hyperparameter configuration of neural networks for any raw tabular dataset. We find the best hyperparameters quickly and consistently while reducing the number of trials required by the tuner to search for optimal hyperparameter configuration. We show encouraging results on OpenML classification datasets.
- **NitroML: Accelerate AutoML Research:** Open-sourced a new benchmarking framework for Automated Machine Learning (AutoML) researchers. NitroML enables AutoML research teams to iterate more quickly on their highly-customized, heterogeneous, and multi-stage ML pipelines. It offers machine learning model-quality benchmarking best practices out-of-the-box, curates raw datasets, and leverages [TFX:OSS] to scale with Cloud-resources. We gave a demo of NitroML at the International Conference on Machine Learning (ICML) 2020. [[Github: google/nitroml](#)]

Indian Institute of Technology (IIT) Kanpur, Project Associate

(2017–2018)

Worked on the following research projects:

- **Generative Model with Bayesian Nonparametric prior for graphs.** Presented at ICML 2019.
- **Scan Registration for Autonomous Driving.** Presented at ICRA 2018.
- **Deep Topic Modeling for Multi-Label Learning.** Presented at AISTATS 2019.

INDUSTRY
EXPERIENCE

• **Software Engineer at Zomato** (June '14 - Jan '16)

- Was a core member of the technical team. Developed server side APIs for various consumer and enterprise products provided by Zomato across 22 countries including India, US and Canada. - Was part of the mobile development team. I was responsible for the development and release cycles of the primary Zomato application which had more than 5 million users. [[link](#)]

OPEN SOURCE
PROJECTS &
INTERNSHIPS

• **Zero-Shot Learning using GANs**

Proposed a wasserstein generative-adversarial-network (W-GAN) to synthesize data for unseen classes conditioned on class attributes as an application in *Zero-Shot Learning*. [[code](#)]

• **Learning Disentangled Representations under Supervision**

Implemented an extension of the vanilla recurrent variational auto-encoder. The architecture had a combination of encoder-generator and a discriminator (trained in wake-sleep fashion) that allowed the network to *learn structured disentangled representations* for text data. [[code](#)]

• **Learning Equivariant Representations**

Implemented a capsule based architecture to *learn equivariant representations* which could be used for pose regression. Open-sourced a proof of concept implementation of “Transforming Autoencoders” to learn transformations. [[code](#)]

• **Research Intern at University of Birmingham, UK** (Dec '13 - Jan '14)

Developed a paradigm and an experimental game in Python to investigate how people’s perception of a hidden partner affect their behavior while competing against an artificial intelligent bot. Used the existing speech recognition model to create an end-end application for voice based commands.

• **Research Intern at University of Leeds, UK** (Jul - Aug '13)

Designed a vision algorithm for cell (lamallopodia) tracking in DIC microscopic videos using entropy differences in subsequent frames. Real-time implementation was further improved by implementing the Extended Kalman filter.

TECHNICAL
SKILLS

- **Deep Learning Frameworks:** Tensorflow, Pytorch.
- **Languages:** Python, CUDA for GPU, MATLAB, PHP, L^AT_EX.
- **Statistical Modeling:** Stan, Edward (Bayesian methods in deep networks).
- **Courses:** Machine Learning - Ph.D. Level, Vector Space with Applications, Text Data Analysis - NLP, Bayesian Methods and Modern Applications, Deep Learning, Applied Stochastic Processes, Learning in Adversarial Setting, Reinforcement Learning at Scale, Causal Inference.

TEACHING
EXPERIENCE

- **Teaching Assistant, ECE 685D: Deep Learning, Duke University**
- **Project Mentor, Probabilistic Machine Learning, IIT Kanpur**
Course Instructor: Dr. Piyush Rai
- **Technical Head, Society of Software Engineering, DTU**

REVIEWER
SERVICE

- International Conference on Machine Learning (ICML)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- International Conference on Artificial Intelligence and Statistics (AISTATS)

GRANTS

- Duke ECE Conference Travel Award for NeurIPS & ICML 2019.
- ICRA Travel Grant.